

GEOGRAPHY

INTRODUCTION

Geography is about determining the significance of “place” as it relates to the natural environment, the human environment, and interactions within and between them. To investigate geographic issues, students must analyse the influences and interrelationships that give a place its distinctive characteristics and thus its spatial importance. Geographic analysis also requires an investigation of the economic, environmental, social, and political perspectives that relate to an issue. The application of the concepts of geographic thinking, spatial skills, and the use of field studies are central to the geographic inquiry process and the learning of geography.

Strands

The Grade 11 and 12 geography courses each have five strands. In all of these courses, strand A, Geographic Inquiry and Skill Development, is followed by four content strands, which are organized thematically.

Citizenship Education

The expectations in the Grade 11 and 12 geography courses provide opportunities for students to explore a number of concepts connected to the citizenship education framework (see page 13).

The Concepts of Geographic Thinking

The four concepts of geographic thinking – spatial significance, patterns and trends, interrelationships, and geographic perspective – underpin thinking and learning in all geography courses in the Canadian and world studies program. At least one concept of geographic thinking is identified as the focus for each overall expectation in the content strands of these courses. The following chart describes each concept and provides sample questions related to it. These questions highlight opportunities for students to apply a specific concept in their studies. (See page 16 for a fuller discussion of the concepts of disciplinary thinking.)

Spatial Significance
This concept requires students to determine the importance of a place or region. They explore the connections that exist between the geographical location and physical characteristics of a site and analyse the unique relationships that exist in and between the natural and human environments in a particular place. Students come to understand that the significance of the same place may be different for humans, animals, and plants. Related Questions* – What does a population settlement map tell us about the preferred range of latitude for settlement in Canada? (CGC1P, B2.2) – How does the official plan for your community address urban sprawl? (CGC1D, E2.3) – How might the uniqueness of a destination make it more appealing to some and less appealing to others? (CGG3O, B1.2) – How does the location of a site influence the feasibility of different types of power generation there? (CGR4E, E3.4) – What are some of the challenges that communities in permafrost regions face in providing safe drinking water and disposing of waste? (CGR4M, E2.3)
Patterns and Trends
This concept requires students to recognize characteristics that are similar and that repeat themselves in a natural or human environment (patterns) and characteristics or traits that exhibit a consistent tendency in a particular setting over a period of time (trends). The characteristics may be spatial, social, economic, physical, or environmental. Students analyse connections between characteristics to determine patterns; they analyse connections between those characteristics over time to determine trends. Related Questions – What trends do you see in the use of alternative energy in Ontario? (CGC1P, A1.5) – What pattern or patterns do you see in the location of First Nations reserves across Canada? (CGC1D, D3.1) – What factors have led to population movement in the region? What specific locations have attracted migrants? Why? (CGD3M, E2.1) – In what ways has technological change amplified the impact of population growth on the natural environment? In what ways has it diminished it? (CGW4U, C2.2)

* These questions are drawn directly from the overview charts that precede the geography courses and from the sample questions that accompany many specific expectations. To highlight the continuity between the geography courses in Grade 11 and 12 and those in Grade 9, and to show possible development in the use of the concepts of geographic thinking over those grades, the chart includes some questions from the Grade 9 geography curriculum as well.

Interrelationships

This concept requires students to explore connections within and between natural and human environments. The interconnected parts of an environment or environments work together to form a system. Students must understand the relationships that exist within a system and then critically analyse the relationships between systems in order to determine the impact they have on one another.

Related Questions

- How does surrounding farmland support a community, and what stresses might the community place on the farmland? (CGC1P, E2.1)
- How might the breaking up of continental ice in Greenland and the Antarctic affect Canada's coastline? (CGC1D, B2.1)
- Why do we need to understand the earth's physical processes in order to predict their potential hazards? (CGF3M, Overview)
- How can international trade policies and corporate actions influence disparities within or between ecumenes? (CGU4M, D1.2)

Geographic Perspective

This concept requires students to consider the environmental, economic, political, and/or social implications of the issues, events, developments, and/or phenomena that they are analysing. In order to solve problems, make decisions or judgements, or formulate plans of action effectively, students need to develop their ability to examine issues from multiple perspectives.

Related Questions

- In what ways can cultural diversity enrich the life of a community? (CGC1P, D1.2)
- Does the financial benefit of extracting natural resources justify related social and/or environmental impacts? (CGC1D, A1.6)
- How might spatial technologies help governments assess the economic, environmental, infrastructural, and social risks of a potential disaster? Why is it important for governments to know and understand these risks? (CGT3O, D1.2)
- Why might some people believe that there is an ethical imperative to share the world's resources equitably? (CGW4C, Overview)
- Why might military and/or government satellite information be shared with or withheld from certain countries or military alliances? What are the implications of doing so? (CGO4M, D2.1)

The Geographic Inquiry Process

In each of the geography courses in the Canadian and world studies curriculum, strand A focuses explicitly on the geographic inquiry process, guiding students in their investigations of issues, events, developments, and/or various geographic phenomena. This process is *not* intended to be applied in a linear manner: students will use the applicable components of the process in the order most appropriate for them and for the task at hand. Although strand A covers all of the components of the inquiry process, it is important to note that students apply skills associated with the inquiry process throughout the content strands in each course. (See page 32 for a fuller discussion of the inquiry process in the Canadian and world studies program.)

The following chart identifies ways in which students may approach each of the components of the geographic inquiry process.

Formulate Questions
Students formulate questions: – to explore various events, developments, issues, and/or phenomena that are related to the overall expectations in order to identify the focus of their inquiry – to help them determine which key concept or concepts of geographic thinking are relevant to their inquiry – that reflect the selected concept(s) of geographic thinking – to develop criteria that they will use in evaluating data, evidence, and/or information; in making judgements, decisions, or predictions; in reaching conclusions; and/or in solving problems
Gather and Organize
Students: – collect relevant qualitative and quantitative data, evidence, and information from field studies^a and a variety of primary and secondary sources,^b including visuals^c and community resources^d – determine if their sources are credible, accurate, and reliable – identify the purpose and intent of each source – identify the points of view in the sources they have gathered – use a variety of methods to organize the data, evidence, and/or information they have gathered – record the sources of the data, evidence, and information they are using – decide whether they have collected enough data, evidence, and/or information for their inquiry

a. Field studies may include, but are not limited to, studies in local neighbourhoods, school grounds, and various sites that allow students to explore different land uses (e.g., recreational, commercial, industrial, and transportation uses).

b. Primary sources may include, but are not limited to, census data, land claims, letters, photographs, speeches, and works of art. Secondary sources may include, but are not limited to, documentaries and other films, news articles, reference books, and most websites.

c. Visuals may include, but are not limited to, satellite images, maps, globes, models, graphs, and diagrams.

d. Community resources may include, but are not limited to, local conservation areas, resources from community groups and associations, government resources, and local plans.

Interpret and Analyse
Students: – analyse data, evidence, and information, applying the relevant concepts of geographic thinking (see preceding chart) – use different types of graphic organizers to help them interpret and/or analyse their data, evidence, and information – identify the key points or ideas in each source – analyse graphs, charts, diagrams, and maps – construct graphs, charts, diagrams, and maps to help them analyse the issue, event, development, or phenomenon they are investigating – analyse their sources to determine the importance of an issue, event, development, or phenomenon for individuals or groups – identify biases in individual sources – determine if all points of view are represented in the source materials as a whole, and which, if any, are missing
Evaluate and Draw Conclusions
Students: – synthesize data, evidence, and information, and make informed, critical judgements based on that data, evidence, and information – determine the short- and long-term impact of an event, development, issue, or phenomenon on people and/or places – reach conclusions about the subject of their inquiry, and support them with their data, evidence, and information – make predictions based on their data, evidence, and information – determine the ethical implications of an issue or action – determine the action required, where appropriate
Communicate
Students: – use appropriate forms (e.g., oral, visual, written, multimedia) for different audiences and purposes – communicate their arguments, conclusions, predictions, and plans of action clearly and logically – use geographical terminology and concepts correctly and effectively – cite sources, using appropriate forms of documentation

Spatial Technologies in Action, Grade 12

University/College Preparation

CGO4M

This course provides a foundation for students who are considering a career involving computer-based spatial technologies. Students will analyse and propose solutions to real-life issues related to spatial organization, such as determining transportation routes, appropriate locations for community services, or potential conservation and preservation areas. Students will extend their ability to use geographic information systems (GIS), global positioning systems (GPS), and remote sensing and to create maps, charts, and graphs. Throughout the course, students will apply the concepts of geographic thinking and the geographic inquiry process to investigate various issues related to spatial organization.

Prerequisite: Any university, university/college, or college preparation course in Canadian and world studies, English, or social sciences and humanities

OVERVIEW

The course has five strands. Instruction and learning related to the expectations in strand A are to be interwoven with instruction and learning related to expectations from the other four strands. Strand A must not be seen as independent of the other strands. Student achievement of the expectations in strand A is to be assessed and evaluated *throughout* the course.

Strand A

A. Geographic Inquiry and Skill Development
Overall Expectations
A1. Geographic Inquiry: use the geographic inquiry process and the concepts of geographic thinking when conducting investigations using spatial technologies
A2. Developing Transferable Skills: apply in everyday contexts skills, including spatial skills, developed through geographical investigation, and identify some careers in which a background in geography might be an asset

(continued)

Overview *(continued)*

Throughout this course, when planning instruction, teachers should weave the expectations from strand A in with the expectations from strands B–E.

Strands B–E

Overall Expectations and Related Concepts of Geographic Thinking	Big Ideas*	Framing Questions*
B. Spatial Organization: Concepts and Processes		
B1. Applying Spatial Technology Skills: demonstrate the ability to use a variety of spatial technologies to collect and analyse data, communicate the results of their investigations, and make decisions based on their analyses (FOCUS ON: <i>Spatial Significance; Geographic Perspective</i>)	Spatial technologies are essential tools for conducting geographic inquiries.	How can you use spatial technologies to communicate the results of an investigation? Why is it important to use a variety of spatial technologies and related data when analysing a geographical issue?
B2. Interpreting Spatial Data: interpret and analyse data produced by a variety of spatial technologies (FOCUS ON: <i>Spatial Significance; Patterns and Trends</i>)	Spatial technologies allow people to analyse a wide variety of visual and numerical data.	What are the limitations of spatial technologies?
B3. Fundamentals of Spatial Organization: describe and apply the fundamental components of spatial organization used in a variety of spatial technologies (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	It is important to understand how various spatial technologies and related data can work together.	How do spatial technologies relate to spatial organization?
C. Using Spatial Technologies to Support Sustainability		
C1. Supporting Sustainable Environments: apply spatial technologies to assess and support the sustainable use of natural and human environments (FOCUS ON: <i>Spatial Significance; Geographic Perspective</i>)	Spatial technologies are key tools in the analysis and assessment of issues related to sustainable use.	How might spatial technologies help various communities and groups assess issues related to sustainability?
C2. Identifying Environmental Patterns and Trends: apply, and explain the role of, spatial technologies in analysing environmental patterns and trends and making decisions related to sustainability (FOCUS ON: <i>Patterns and Trends; Interrelationships</i>)	Spatial technologies are key tools in the analysis of interactions between human activity and physical features and processes.	Why is it important to use data and information from a number of different spatial technology sources when analysing issues?

* See page 17 for a discussion of the purpose of big ideas and framing questions.

Overall Expectations and Related Concepts of Geographic Thinking	Big Ideas*	Framing Questions*
D. Spatial Systems, Interconnections, and Interdependence		
D1. Relationships between Systems: investigate and analyse relationships and interactions within and between physical and human systems, using a variety of spatial technologies (FOCUS ON: <i>Spatial Significance; Interrelationships</i>)	Interactions within and between systems should be analysed in order to assess their impacts and possible risks.	How can information and data from various spatial technologies be used for risk assessment and preparation?
D2. International and Social Implications: investigate and evaluate the role of spatial technologies in national and global security and safety, global communications, and international cooperation (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	The use of spatial technologies to collect data and information is beneficial but also can be ethically problematic.	Under what circumstances might the use of spatial technologies be a source of disagreements or conflict?
D3. Global System Interactions: apply, and explain the use of, spatial technologies to analyse the interactions between people, places, and issues at a global level (FOCUS ON: <i>Patterns and Trends; Interrelationships</i>)	Spatial technologies are key tools in the analysis of global interactions.	How might two distinct groups use spatial technologies for different purposes? What are the ethics of data collection using spatial technologies?
E. Dynamic Impacts within Communities		
E1. Monitoring and Predicting Change: apply, and assess the use of, spatial technologies to monitor, model, and predict change in human and physical environments at the local, national, and global levels (FOCUS ON: <i>Spatial Significance; Patterns and Trends</i>)	Spatial technologies allow people to study how and why change occurs and to use this understanding to make predictions.	What can you learn from conducting a longitudinal study of natural and human change at a local, national, or global level? Who might benefit from this information?
E2. Development of Spatial Technologies: describe developments in spatial technologies over time and potential developments in the future, and analyse how these developments affect the uses and users of these technologies (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	Spatial technology is being used in almost every type of workplace.	How do spatial technologies aid in making predictions about future events or disasters? How have the purpose and design of spatial technologies changed over time? Why have these changes occurred?

A. GEOGRAPHIC INQUIRY AND SKILL DEVELOPMENT

OVERALL EXPECTATIONS

Throughout this course, students will:

- A1. Geographic Inquiry:** use the geographic inquiry process and the concepts of geographic thinking when conducting investigations using spatial technologies;
- A2. Developing Transferable Skills:** apply in everyday contexts skills, including spatial skills, developed through geographical investigation, and identify some careers in which a background in geography might be an asset.

SPECIFIC EXPECTATIONS

A1. Geographic Inquiry

Throughout this course, students will:

- A1.1** formulate different types of questions to guide investigations using spatial technologies (e.g., factual questions: *How is vegetation health identified on an infrared photograph?*; comparative questions: *Which map scale is most effective for identifying a transportation route?*; causal questions: *What factors might cause the loss or degradation of a GPS signal?*)

- A1.2** select and organize relevant data and information on geographic issues from a variety of primary and secondary sources (e.g., primary: *raw data from fieldwork, both quantitative and qualitative; photographs; satellite images*; secondary: *published statistics, newspapers, books, atlases, geographic magazines, websites, graphs, charts, digital and print maps*), ensuring that their sources represent a diverse range of perspectives

Sample questions: “How might you use quality of life indicators as layers on a GIS map to analyse poverty distribution patterns in different parts of the world? Where might you find this data and information?” “What type of data and information do you need to collect in order to assess the impact of a tsunami on a specific country and people?” “What types of maps and graphs will help you analyse the environmental impact of an urban development project?”

- A1.3** assess the credibility of sources and information relevant to their investigations (e.g., *by considering how the data are constructed to support*

the author’s point of view, the possible bias of the author, the expertise of the author, the accuracy of the text and supporting data, the intended audience, the purpose of the messaging, the context in which the information was presented)

Sample questions: “What is the source of this map or photographic image? What biases might this source have and how might they affect the map or image? Have you consulted other maps or images of the same place from a different source or a slightly different time period? How do they compare?”

- A1.4** interpret and analyse data and information relevant to their investigations, using various tools, strategies, and approaches appropriate for geographic inquiry (e.g., *interpret data related to the release of a specific pollutant from various point sources; analyse navigational graphs and charts for selected locations to determine the best route for travel; use decision-making templates to analyse points of view on an issue; use a graphic organizer to outline the pros and cons of various map projections, and choose the best one for their purposes*)

Sample questions: “What type of graphic organizer would you use to help you interpret data gathered from a GPS?” “What data layers and types of information might you include when using GIS in order to analyse statistics on the various ethnic communities of a city?”

- A1.5** use the concepts of geographic thinking (i.e., spatial significance, patterns and trends, interrelationships, geographic perspective) when analysing and evaluating data and information,

formulating conclusions, and making judgements about issues they are investigating through the use of spatial technologies (e.g., *use the concept of spatial significance to analyse an area of urban sprawl; use the concept of patterns and trends to analyse short- and long-term graphs of ocean temperatures and determine how they have changed over time; use the concept of interrelationships to make correlations between urban areas and ground-level ozone pollution or the number of smog days in an area; use the concept of geographic perspective to analyse social, political, economic, and environmental impacts of the use of GPS technology*)

Sample questions: “How can an understanding of spatial significance help you when determining where a natural area or green belt should be located?” “How might an understanding of patterns and trends help you analyse the impact on coastlines of increased precipitation in temperate zones?” “How might an understanding of interrelationships guide your analysis of a story map showing the connections among the following: a product you buy, the company that sells it, the resources used to make and dispose of it, and labour conditions in the company that manufactures it?” “How can geographic perspective help you identify data layers needed to analyse the impacts of globalization on indigenous peoples?”

- A1.6** evaluate and synthesize their findings to formulate conclusions and/or make informed judgements or predictions about the issues they are investigating

Sample questions: “What did you find out about the relationships among oil and gas pipeline routes, land claims, and animal migration pathways? In what ways might these relationships continue to be an issue in the future, such as if migration pathways shift or new pipelines are laid? What conclusions can you draw about the best locations for future pipeline routes?”

- A1.7** communicate their ideas, arguments, and conclusions using various formats and styles, as appropriate for the audience and purpose (e.g., *a debate for classmates on the best scale interval to use in constructing a map for a particular message; a video for a Grade 9 class illustrating how colour is used on land-use maps and how shading is used on choropleth maps to communicate intended messages; a webcast or podcast for the general public on secure data management strategies for GPS-enabled devices such as cellphones; an annotated map using photos illustrating the landmark features and points of interest for a particular site [city, conservation area, or park] to be used in the tourism industry*)

Sample questions: “What kind of information does your audience need?” “What symbols or shading techniques would best communicate the intended message?” “What format and approach would be most effective in conveying your information to this particular audience?”

- A1.8** use accepted forms of documentation (e.g., footnotes, author/date citations, reference lists, bibliographies, annotated bibliographies, credits) to reference different types of sources (e.g., websites, blogs, books, articles, films, data)

- A1.9** use appropriate terminology when communicating the results of their investigations (e.g., *vocabulary specific to their inquiry; terminology related to geography and to the concepts of geographic thinking*)

A2. Developing Transferable Skills

Throughout this course, students will:

- A2.1** describe ways in which geographic investigation can help them develop skills, including spatial skills and the essential skills in the Ontario Skills Passport (e.g., *reading graphic texts, writing, graphing, computer use, use of spatial technologies, oral communication, numeracy, decision making, planning, management, finding information, problem solving*), that can be transferred to postsecondary opportunities, the world of work, and everyday life

- A2.2** apply in everyday contexts skills and work habits developed through geographic investigation (e.g., *ask questions to deepen their understanding of a global issue; use quantitative data to support an idea; use spatial skills to determine relationships between regions of the world; apply work habits such as collaboration to help them make a decision*)

- A2.3** apply the concepts of geographic thinking when analysing current events involving geographic issues (e.g., *use the concept of spatial significance to analyse possible reasons for the extirpation or extinction of plant and animal species; use the concept of patterns and trends to analyse when major earthquakes might occur on the west coast of North America; use the concept of interrelationships to analyse the connection between the percentage of potable water in an area and the prevalence in that area of diseases associated with sanitary conditions; use the concept of geographic perspective to analyse the impact of climate change in various regions of the world*) in order to enhance their understanding of these issues and their role as informed citizens

A2.4 identify some careers in which a geography background might be an asset (*e.g., cartographer, border services officer, civil engineer, transportation planner, emergency planning specialist, military personnel, real estate agent, land surveyor, marketing analyst, town planner, GIS technician*)

B. SPATIAL ORGANIZATION: CONCEPTS AND PROCESSES

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1. Applying Spatial Technology Skills:** demonstrate the ability to use a variety of spatial technologies to collect and analyse data, communicate the results of their investigations, and make decisions based on their analyses (**FOCUS ON:** *Spatial Significance; Geographic Perspective*)
- B2. Interpreting Spatial Data:** interpret and analyse data produced by a variety of spatial technologies (**FOCUS ON:** *Spatial Significance; Patterns and Trends*)
- B3. Fundamentals of Spatial Organization:** describe and apply the fundamental components of spatial organization used in a variety of spatial technologies (**FOCUS ON:** *Interrelationships; Geographic Perspective*)

SPECIFIC EXPECTATIONS

B1. Applying Spatial Technology Skills

FOCUS ON: *Spatial Significance; Geographic Perspective*

By the end of this course, students will:

- B1.1** use a variety of spatial technologies to gather data through a field study or survey, code and tabulate the data (*e.g., by performing tasks such as gathering data through GPS coordinates and attributes of points, lines, and polygons; buffering data; geocoding; georeferencing; creating new spreadsheet fields; building attribute tables*), and interpret the results to identify patterns in the data

Using spatial skills: In a field study, students can measure wind velocity in an urban centre and map the data in GIS to determine the pattern and identify the areas that experience the greatest wind chill. Students can use GPS to determine the elevation parameters needed to level agricultural fields in order to create an efficient irrigation system.

- B1.2** query spatial databases they have built (*e.g., request specific features or tabular attributes*) to answer practical research questions

Sample question: “Which features or tabular data would you need to request to determine the interrelationship between the demographics of a region and voter turnout?”

Using spatial skills: Students can query their databases to request specific features or tabular data to determine the best transportation route for agricultural goods or the impact of disease on local vegetation.

- B1.3** use spatial technologies to analyse geographic case studies and make associated decisions (*e.g., determine the best route for a road or trail, the best location for a building, or the best place for a specific land use; determine the area of impact of a physical phenomenon or disaster*)

Sample questions: “What is the best route for a cycling, running, or snowmobile competition in your local area? How do you know?” “How large are the deposits of a valuable natural resource in a particular country?” “How would traffic flow be changed by adding a turning lane to a particular street?” “What is the best route for a recreational trail in an environmentally sensitive area?”

- B1.4** construct a map that communicates qualitative and/or quantitative attribute data, using mapping conventions (*e.g., title, scale, legend, orientation*) as appropriate

Using spatial skills: Students can generate maps in GIS using attribute values such as physical data (*e.g., climate, drainage features, ecozones*) and human data (*e.g., population density, ethnicity, voter turnout*). Different

types of symbology (e.g., graduated colour, proportional symbols) and classification (e.g., natural breaks, equal intervals, quantiles) can be added to a legend. Different scale intervals can be explored to determine the most meaningful way to present the information.

B2. Interpreting Spatial Data

FOCUS ON: *Spatial Significance; Patterns and Trends*

By the end of this course, students will:

B2.1 interpret a variety of images (e.g., *aerial photographs, satellite images, web-based map images*) of different areas of the world, describing observed spatial characteristics (e.g., *tone, texture, shape, pattern, size, association, shadow*) and identifying physical and human features based on direct observations or inferences

Sample questions: “What is the difference between observation and inference?” “How do you use inference when reviewing different types of images?”

B2.2 interpret satellite images (e.g., *remote sensing images, infrared images*) and use their conclusions to organize major features of the world (e.g., *mountain systems, vegetation belts, oceans*) into spatial regions

B2.3 interpret a variety of maps (e.g., *thematic, topographic*), aerial photographs, and satellite images to analyse patterns of physical and human features

Sample question: “How do rivers, glaciers, trees, and buildings appear on a topographic map, an aerial photograph, and a satellite remote sensing image?”

Using spatial skills: Students can use different types of base maps (e.g., topographic imagery, terrain) in a GIS to explore and analyse the various ways in which physical landforms, drainage, vegetation and human patterns, transportation networks, and population distribution may be represented (e.g., through the use of contour lines, shading, colour).

B3. Fundamentals of Spatial Organization

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

B3.1 describe and calculate the ways in which various types of scales and resolutions in cartography, GIS, remote sensing, and GPS

(e.g., *small scale, large scale, linear scale, statement, representative fraction, number of pixels, map resolution, spatial resolution*) affect the degree of clarity of a map or image

Sample question: “How does the number of pixels influence the clarity of an image at various scales?”

B3.2 explain direction as used in cartography, GIS, remote sensing, and GPS, including, but not limited to, the following concepts: true north, magnetic north, grid directions, bearings, azimuth, nadir, “look direction”, range, total field of view, and flight path

Sample questions: “How does the earth’s shape affect the placement of the direction arrow on a map?” “How do the earth’s lines of latitude relate to distance?” “Why is it important to understand the concept of a great circle route when determining a flight path?”

B3.3 apply the concept of location in cartography, GIS, remote sensing, and GPS, using geographic coordinates (e.g., *latitude and longitude, the Universal Transverse Mercator [UTM] system, geodetic datum*), geocoding (e.g., *street address, postal code*), georeferencing, geocaching, routing, and triangulation and trilateration

B3.4 describe key concepts associated with elevation in cartography (e.g., *spot elevation, contour lines, shading and grading, benchmarks*), GIS, remote sensing, land surveying, and digital elevation models (DEMs)

B3.5 identify the properties and uses of different types of map projections (e.g., *azimuthal, conical, cylindrical, equal-area cylindrical, conformal, gnomonic, stereographic, oblique*)

Sample questions: “How do different projections in GIS software affect the way in which various countries or continents are presented?” “Why would someone choose one projection over another?”

Using spatial skills: Students can use GIS to compare the scale of distances between features and the size of the total area for the same region of the world using various map projections.

B3.6 describe various means of acquiring information and data about the earth’s surface using cartography, GIS, remote sensing, and GPS (e.g., *the process involving the energy source, its structure, and the interactions with the atmosphere and the target; the organization of the electromagnetic spectrum, active versus passive sensing, types of platforms [satellites, airplanes, handheld cameras], orientations of platforms [orbits, paths], types of*

sensors [multispectral scanner, radar, digital camera], receiving stations, manual and scanning digitization)

Sample questions: “What types of information can be gathered using remote sensing? How does that compare with other types of imaging tools?” “In what circumstances might one use lidar instead of radar?”

B3.7 explain the different types and characteristics of data needed to create an image or construct a map (e.g., types: *analogue versus digital data, sets of layers or themes, spatial versus non-spatial data and the concept of attributes, qualitative and quantitative data, raster versus vector data*; characteristics: *accuracy; pixel value; the matrix structure of an image versus points, lines, and polygons of GIS; spectral signatures and spectral reflectance curves such as histograms; various types of image enhancements; classification techniques such as supervised versus unsupervised data*)

C. USING SPATIAL TECHNOLOGIES TO SUPPORT SUSTAINABILITY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1. Supporting Sustainable Environments:** apply spatial technologies to assess and support the sustainable use of natural and human environments (**FOCUS ON:** *Spatial Significance; Geographic Perspective*)
- C2. Identifying Environmental Patterns and Trends:** apply, and explain the role of, spatial technologies in analysing environmental patterns and trends and making decisions related to sustainability (**FOCUS ON:** *Patterns and Trends; Interrelationships*)

SPECIFIC EXPECTATIONS

C1. Supporting Sustainable Environments

FOCUS ON: *Spatial Significance; Geographic Perspective*

By the end of this course, students will:

- C1.1** perform assessments of the natural environment (*e.g., assessments of wildlife or of wetlands*), using spatial technologies as appropriate, to analyse sustainability

Sample questions: “Which spatial technologies would you use to do wildlife assessments such as determining the levels of fish stocks or tracking caribou migration patterns?” “How do spatial technologies help to map and predict the extent of the following: an oil spill; the effect of disease outbreaks on agricultural crops or forests; deforestation and the effect of commercial logging operations?” “How might spatial technologies help to map and monitor the destruction of wetlands?”

Using spatial skills: Students can use GIS to explore data using time attributes to view changes in a characteristic for a given area, such as the change in carbon dioxide in the global atmosphere over time or the change in forest cover for a particular area. Students can use proximity and overlay tools in GIS to do a susceptibility study to determine the likelihood of changes to the natural environment such as the spread of insects, or the likelihood of a natural disaster such as a forest fire or landslide.

- C1.2** use spatial technologies to construct maps to support sustainable human activities (*e.g., to*

show the most sustainable route for a hiking or cycling path, to show the best route for a road through an environmentally sensitive region, to determine the best location for infrastructure such as a pipeline or power line, to show the impact of increased human activity in a park)

Sample questions: “How would you construct a map showing the impacts of increased human activity on local, provincial, and national parks?” “What information would you need to plan a sustainable hiking or cycling path using web-based GIS, a digital elevation model [DEM], or GPS?” “How might spatial technologies help in planning and creating buffer zones near open pit mines, quarries, and tailing ponds?”

Using spatial skills: Students can use GIS to analyse information such as vector data (points, lines, and polygons) and raster data (pixel images, satellite images, and aerial images) in order to create buffers or zones delineating sensitive environmental areas. Students can use a digital elevation model to determine where best to situate a ski resort.

- C1.3** assess issues of sustainability (*e.g., environmental, social, economic, political issues*) relating to First Nations, Métis, and Inuit peoples and communities (*e.g., issues relating to development projects such as the construction of hydroelectric dams, oil and gas extraction processes*), using spatial technologies as appropriate

Sample questions: “What data layers illustrate the impact on First Nations, Métis, and Inuit communities of the decline of traditional food

sources due to environmental degradation?" "How might land claim issues have an impact on resource development?" "How does the use of spatial technologies support groups using the First Nations Environmental Assessment Toolkit?" "How might First Nations, Métis, and Inuit communities be affected by water quality, garbage disposal, and sewage treatment issues?"

C2. Identifying Environmental Patterns and Trends

FOCUS ON: *Patterns and Trends; Interrelationships*

By the end of this course, students will:

C2.1 analyse interactions between specific physical features and processes and human features and activities, and explore issues of sustainability related to these interactions, using a variety of spatial technologies

Sample questions: "Where are landslides or coastal erosion likely to occur? How might human activity contribute to or mitigate these hazards?" "How do spatial technologies help detect and monitor forest-fire risk areas, water quality, invasion of foreign species, and dust storms? How can this information be used to increase sustainability?" "What data layers might be used to illustrate how large recreational facilities such as golf courses, ski runs, and amusement parks might affect the use of prime agricultural land nearby?"

C2.2 analyse the changing relationships between patterns of physical features and patterns of human activity (e.g., *the relationship between the spatial distribution of arable land and the location of large urban settlements*), and determine the implications for sustainability, using spatial technologies

Sample questions: "How do spatial technologies help to identify landscape features, the different processes of their formation, and their changes over time?" "How do spatial technologies help to assess the location of settlements in terms of sustainability?" "How do spatial technologies help us assess the ways in which the necessity of feeding growing populations is affecting the environment?" "How do spatial technologies aid in our understanding of issues around land reclamation and the flooding of New Orleans during Hurricane Katrina? How can we identify other reclaimed land that might be vulnerable to flooding or other physical processes?"

D. SPATIAL SYSTEMS, INTERCONNECTIONS, AND INTERDEPENDENCE

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1. Relationships between Systems:** investigate and analyse relationships and interactions within and between physical and human systems, using a variety of spatial technologies (**FOCUS ON:** *Spatial Significance; Interrelationships*)
- D2. International and Social Implications:** investigate and evaluate the role of spatial technologies in national and global security and safety, global communications, and international cooperation (**FOCUS ON:** *Interrelationships; Geographic Perspective*)
- D3. Global System Interactions:** apply, and explain the use of, spatial technologies to analyse interactions between people, places, and issues at a global level (**FOCUS ON:** *Patterns and Trends; Interrelationships*)

SPECIFIC EXPECTATIONS

D1. Relationships between Systems

FOCUS ON: *Spatial Significance; Interrelationships*

By the end of this course, students will:

- D1.1** use spatial technologies to analyse and illustrate specific relationships within and between natural and human systems

Sample questions: “Is there a correlation between the locations where specific crops are grown and the amount of precipitation at those locations?” “What are the densely populated areas that are most at risk for volcanic and/or seismic disaster?” “How does relief affect drainage systems?”

Using spatial skills: Students can use GIS to create queries to uncover and explore relationships between various phenomena. For example, to create queries analysing where to install solar panels, students could layer spatial data with attribute data indicating the amount of sunlight needed to support the use of a solar panel.

- D1.2** apply a variety of spatial technologies to investigate the physical impact of natural events

and processes that change the physical environment (e.g., *tectonic forces, weathering and erosion, rising sea levels, melting of glaciers, wind systems*)

Sample questions: “How do spatial technologies measure and monitor the extent of flooding and the physical impact on affected areas?” “How do spatial technologies track the paths and impacts of hurricanes, tornadoes, and other major storms?” “What impact do changes in glacial ice in mountain areas have on watersheds downstream?”

- D1.3** use a variety of spatial technologies to assess the potential risks and impacts of interactions between natural and human systems (e.g., *risks associated with using the fertile soil at the base of an active volcano for coffee plantations, the possible impacts of rising sea levels on specific urban centres, the impact of melting sea ice in the Arctic on possible transportation routes*)

Sample questions: “How do spatial technologies map the frequency of earthquakes and volcanoes? Why would this information be helpful in assessing the potential risks of a high-magnitude earthquake and possible tsunami near a densely populated area?”

D2. International and Social Implications

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

D2.1 analyse the ways in which spatial technologies are used in relation to national and global security and safety and international cooperation, and explain some ethical issues that arise from such uses (*e.g., national security versus individual privacy rights*)

Sample questions: “How has the change in U.S. government regulations around the accuracy of GPS systems affected the use of navigational systems?” “Why might military and/or government satellite information be shared with or withheld from certain countries or military alliances? What are the implications of doing so?” “How might disclosures of secret documents and spying influence future development and/or regulation of spatial technologies?”

D2.2 analyse the importance and implications of international cooperation in the sharing and maintenance of global communications and navigation networks (*e.g., issues of control versus openness, protection from terrorism or criminal activity versus individual privacy rights and freedoms*)

Sample questions: “In what ways do countries need to cooperate when it comes to navigation and flight paths? What might happen if these countries did not cooperate and maintain these networks?”

D2.3 describe and assess the positive and negative effects of the use of spatial technologies in international affairs (*e.g., in monitoring and addressing environmental and human issues such as climate change, acid rain, population movements, politics, trade, consumerism, hunger and poverty, disease*)

Sample questions: “How can spatial technologies be used to develop and analyse hypothetical disaster scenarios?” “How can spatial technologies be used to develop disaster response and evacuation plans? What are the limitations of using spatial technologies to support disaster relief, recovery, and relocation operations?” “What are the implications of using GPS to locate and target military objectives?” “How might spatial technologies be used for political ends?”

D2.4 analyse the use of maps and spatial images to advance public- and private-sector interests around the world (*e.g., government propaganda, advertising*), and explain the ways in which bias

is shown in these maps and images (*e.g., choice of map projection or scale to advance a point of view*)

Sample question: “How do the media show bias through their choice of map projections?”

D3. Global System Interactions

FOCUS ON: *Patterns and Trends; Interrelationships*

By the end of this course, students will:

D3.1 describe how spatial technologies are used to identify system connections on a global scale (*e.g., the role of spatial technologies in mapping migration routes, identifying areas of high urban density, and defining areas where military conflict is occurring*), and apply these technologies to identify such connections

Sample question: “How do spatial technologies help to analyse global climate change indicators and outline the areas affected by climate change?”

Using spatial skills: Students can experiment with various population statistical intervals to determine which interval best illustrates urban density on a global scale. Other data layers can be added to investigate possible correlations between urban density and various quality of life indicators.

D3.2 describe ways in which raster data from satellites (*e.g., radar and Landsat images*) are used to provide worldwide information about the earth’s surface (*e.g., tracking of disasters such as oil spills, forest fires, floods, ship collisions; Arctic monitoring activities such as daily ice charts, tracking of iceberg movements; coastal surveillance and tracking information such as shipping routes, offshore oil exploration, coastal erosion, productive fishing zones, illegal fishing activities*)

Sample questions: “How does radar imagery help to monitor disaster zones, assist in emergency response activities, and provide early warning of potential disasters? How does radar imagery help recovery efforts in a disaster zone, such as efforts related to damage assessment, analysis of environmental impacts, reconstruction, and/or rehabilitation?” “What are the benefits of radar compared to other satellite applications and capabilities?”

Using spatial skills: Students can use a selected radar image and add symbols in order to analyse large-scale changes over time, such as differences in the extent of ice cover in Greenland, the area permanently flooded by the construction of the Three Gorges Dam, or areas experiencing impacts from volcanic activity in Iceland.

E. DYNAMIC IMPACTS WITHIN COMMUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1. Monitoring and Predicting Change:** apply, and assess the use of, spatial technologies to monitor, model, and predict change in human and physical environments at the local, national, and global levels (**FOCUS ON:** *Spatial Significance; Patterns and Trends*)
- E2. Development of Spatial Technologies:** describe developments in spatial technologies over time and potential developments in the future, and analyse how these developments affect the uses and users of these technologies (**FOCUS ON:** *Interrelationships; Geographic Perspective*)

SPECIFIC EXPECTATIONS

E1. Monitoring and Predicting Change

FOCUS ON: *Spatial Significance; Patterns and Trends*

By the end of this course, students will:

- E1.1** apply, and assess the use of, spatial technologies, including satellite imagery, to monitor changing features or phenomena and to model and predict future processes and events (*e.g., physical changes shown in a temporal sequence of maps or aerial photographs*)

Sample questions: “How do spatial technologies aid in the study of the effect of climate change in the Arctic? What patterns of change can you identify? What are some of the consequences of these changes for polar species that depend on sea ice for survival?” “In what ways does the use of satellite images of watersheds and ground data of algal blooms in local rivers help us predict the future effects of aquatic death zones?” “How do spatial technologies help in analysing the rate and impact of urban sprawl?”

- E1.2** apply, and assess the use of, spatial technologies to predict long-term change

Sample questions: “What are some of the likely long-term effects of desertification in areas around the edges of the Sahel or the Gobi Desert? How might this phenomenon affect countries in Africa and Asia?” “Using satellite images and GIS, how might you determine how the climate in different regions of the world might be affected as average surface temperatures on the earth continue to rise? What are the likely

long-term impacts of climate change on sea levels, water levels in lakes and rivers, or the extent of permafrost?” “How will the rate of urban sprawl affect productive soil and agricultural land?”

- E1.3** apply, and assess the use of, spatial technologies to model and predict catastrophic events

Sample questions: “Why can the depth of the water along ocean shorelines be an indicator of potential damage by tsunamis on beaches, low-lying settlements, and resort areas?” “Using snow cover data, how can you determine the potential risk of avalanches in mountain areas?” “How can satellite images be used to predict future volcanic activity and improve the safety of surrounding communities or of scientists on site?”

Using spatial skills: Students can use National Oceanic and Atmospheric Administration (NOAA) data and satellite images to predict the path of a developing hurricane or typhoon. Students can use satellite images or the buffering or query techniques in GIS to determine where the highest level of potential damage may occur along a fault line.

- E1.4** use spatial technologies to predict the impact of a possible future change on their own or another community (*e.g., the impact of an increase in the number and severity of storms due to climate change, the impact of urban growth on the local watershed and water resources*)

Sample question: “How can spatial technologies help predict the impacts of climate change on crop growth in your region?”

E2. Development of Spatial Technologies

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

- E2.1** describe the development of satellite technology, and explain some of the ways it has changed daily life and how business is conducted in various economic sectors
- Sample questions:* “What are the different numbers, types, and functions of satellites that have been used by different countries?” “How are various countries using satellites and their data? How has their usage changed over time?” “What are the implications of the increasing numbers of satellites in low Earth orbit?”
- E2.2** identify key Canadian contributions to the development of spatial technologies over time (e.g., RADARSAT-1 and -2, the Alouette and Anik satellites, and other Canadian satellites that are in development or that have been launched with international collaboration; the Canada Centre for Remote Sensing [CCRS]; Canadian Space Agency [CSA] contributions such as Canadarm and the David Florida Laboratory; the work of Canadian astronauts; the development of the Prince Albert Satellite Station; the Brewer Ozone Spectrophotometer; various contributions by Land Information Ontario [LIO] and Natural Resources Canada [NRCan])

- E2.3** assess whether modern spatial technologies could have averted or diminished the consequences of famous historical disasters (e.g., the sinking of the Titanic, the sinking of the Ocean Ranger drilling platform, the eruption of Mount St. Helens, the Frank Slide or the Saint Jean Vianney Slide, the 1883 eruption of Krakatoa, the 1953 North Sea flood)
- E2.4** analyse ways in which advances in spatial technologies affect the users and the use of these technologies (e.g., changes in costs, training requirements, ease of use; types of jobs and careers that use these technologies)
- E2.5** identify and describe the possibilities for using spatial technology applications in non-traditional domains in the future
- Sample questions:* “Why might spatial technologies be an effective tool in combatting crime or criminal activity?” “How could spatial technologies be used in marketing analysis and predicting future needs?” “How could spatial technologies be used more widely in monitoring disease spread and disease control response time?” “How could spatial technologies be used to assess and predict recreational needs in a developing community?”